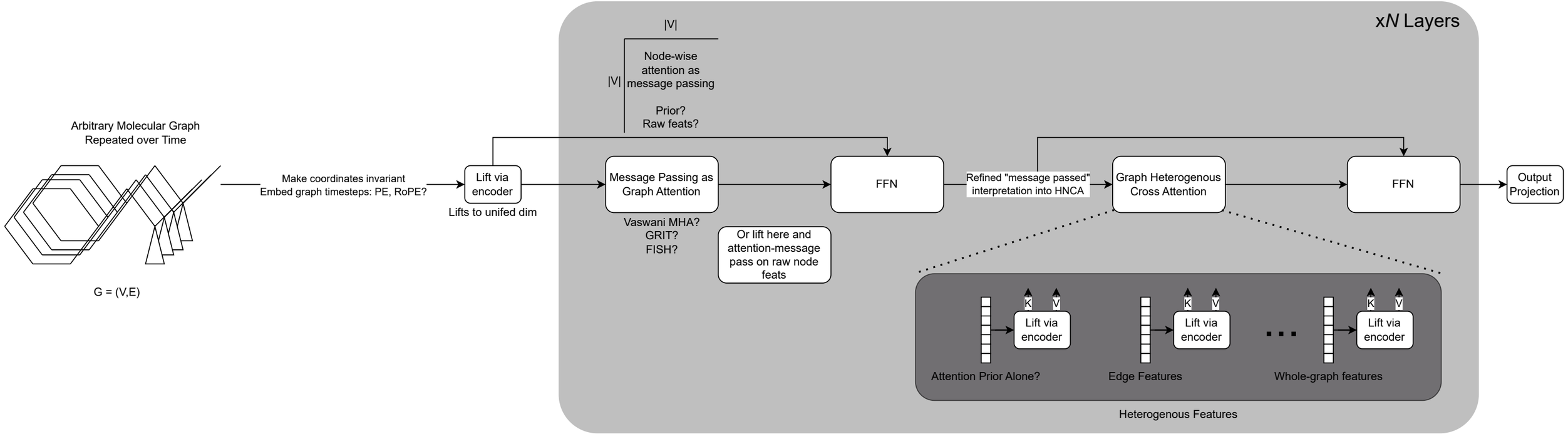
xN Layers
Arbitrary Molecular Graph
Repeated over Time
G = (V,E)
Make coordinates invariant
Embed graph timesteps: PE, RoPE?
Lift via encoder
Lifts to unifed dim
|V|
|V|
Node-wise attention as message passing
Prior?
Raw feats?
Message Passing as Graph Attention
Vaswani MHA?
GRIT?
FISH?
Or lift here and attention-message pass on raw node feats
FFN
Refined "message passed" interpretation into HNCA
Graph Heterogenous Cross Attention
K
V
Lift via encoder
Attention Prior Alone?
Edge Features
Whole-graph features
Heterogenous Features
FFN
Output Projection